How to Solder a JST Connector: Through-Hole, SMT, and Board-to-Board Guide

Published Oct 30, 2026, updated Oct 30, 2026 14 min

Table of Contents

Quick Answer: How Do You Solder a JST Connector?
Know What You're Soldering: JST Series, Pitch, and Mount Type
What You Need: Tools & Materials
How to Solder a Through-Hole JST Connector: Step by Step
How to Solder an SMT JST Connector
Should You Solder or Crimp JST Wires?
How to Solder Board-to-Board JST Connectors

Write a message

Key Takeaways

Soldering a connector is electrical and mechanical at once: Seat the header flush and square, tack one corner pin, verify alignment, then solder the remaining pins and every mechanical anchor tab, because plugging and cable forces travel through the housing and hold-downs rather than the signal pins.

Match the method to the mount type: Through-hole headers need full barrel fill with a larger tip and flux for heat-sinking power pins, SMT headers need tack-and-align with hold-down tabs fully soldered, board-to-board parts need coplanarity checks, and wire-side JST housings are designed for crimps rather than solder.

Reflow only where the part allows: Many SMT JST headers are reflow-rated and pin-in-paste can solder through-hole connectors in the same cycle as the SMD parts, but always confirm the

exact part number's reflow conditions in the datasheet and control paste volume with a proper stencil.

JST connectors may look simple, but achieving reliable solder joints can be more difficult than expected. A header may sit crooked, a large power pin may form a cold joint, or an SMT connector may look electrically fine yet tear from the board after repeated plugging. If you are learning how to solder JST connector, the key is to think about both electrical continuity and mechanical strength.

Connectors experience cable pull, mating force, and repeated insertion, so full through-hole fill, square alignment, and soldered hold-down tabs matter as much as the signal pins. This guide covers through-hole and SMT JST headers, board-to-board connector soldering, crimp versus solder on the wire side, and reflow or pin-in-paste options.

Quick Answer: How Do You Solder a JST Connector?

Seat the connector flush and square, solder one corner pin first, verify alignment, then solder the remaining pins and all mechanical anchor tabs. Use flux and a larger tip for heat-sinking power pins. For SMT JST headers, tack one pad, align the connector carefully, then finish the remaining pads and hold-down tabs.

Know What You're Soldering: JST Series, Pitch, and Mount Type

JST is a connector manufacturer, not a single connector style. Its catalog includes many series such as PH, XH, SH, and GH, with different pitches, current ratings, mounting styles, and mating arrangements. For example, JST lists PH as a 2.0 mm-pitch wire-to-board family and XH as a 2.5 mm-pitch family, with both through-hole and SMT board-mount options available in parts of those ranges.

Write a message

This article focuses on soldering the board-side header and board-to-board connector. The wire side of most JST wire-to-board systems uses a housing with crimp contacts, so proper crimping is the intended method. For series selection, pitch details, and housing/contact matching, see JLCPCB's complete guide to JST connectors and their types. Always confirm the exact footprint and soldering or reflow conditions in the connector datasheet before assembly.

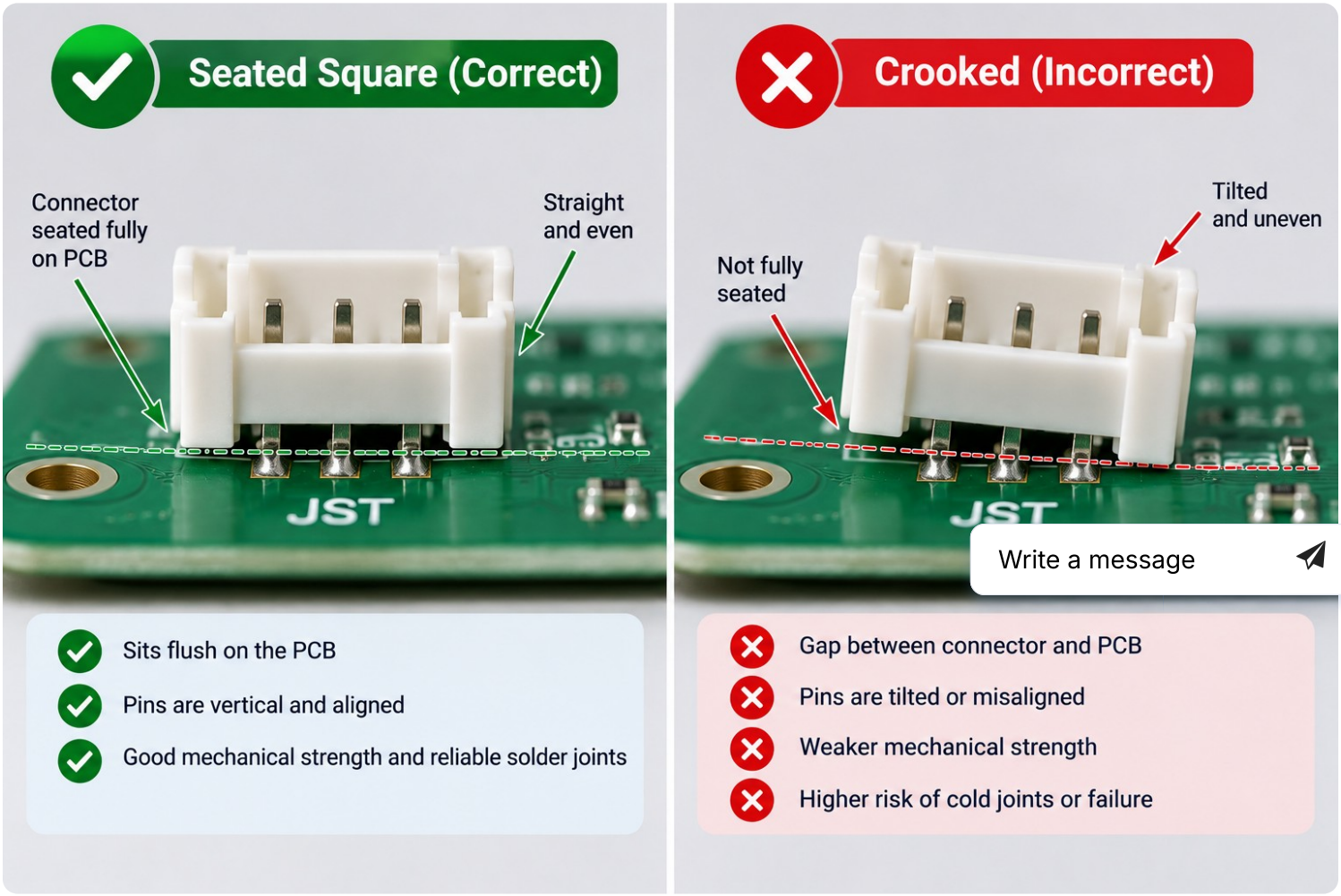
JST publishes a product page for each connector family, listing pitch, current rating, and the available through-hole and SMT board-mount options. The PH and XH pages are the ones you will reach for most often.

What You Need: Tools & Materials

- Temperature-controlled soldering iron with a fine-to-medium tip; use a slightly larger tip for power or ground pins that sink heat.

- Fine solder suitable for your assembly process and a quality flux for fast, clean wetting.
- Tweezers, flush cutters, a PCB holder, helping hands, or a small fixture to keep the connector square.
- Solder wick and/or a desoldering pump for clearing bridges or correcting a crooked pin.
- Isopropyl alcohol (IPA) and a suitable brush or swab for post-solder cleanup where appropriate.
- Optional for SMT or production work: hot air, solder paste, hot plate/reflow oven, and a stencil.

If you want a deeper look at flux behavior on connector pins, see how to use solder flux for through-hole soldering.



AI-generated

How to Solder a Through-Hole JST Connector: Step by Step

01 Insert the header flush and square

Insert the board-side header completely into the PCB footprint and make sure it sits flat. If the connector tends to move or lift when you turn the board over, hold it in place with tape, a fixture, or helping hands.

02 Solder one corner pin only

Use a small amount of solder to tack a corner pin. This single joint holds the connector while still allowing easy correction.

03 Check alignment before continuing

Look from the side and top. If the housing is tilted or floating above the PCB, reheat the tacked pin and press the connector flat before the solder solidifies.

04 Solder the remaining signal and power pins

Heat both the pin and plated pad, then feed enough solder to form a smooth fillet. Large power or ground pins may need a larger tip and slightly longer contact because connected copper planes pull heat away.

05 Solder every mechanical anchor or hold-down tab

These tabs carry insertion and cable forces. Skipping them can leave the connector electrically functional but mechanically weak.

06 Add flux to stubborn joints

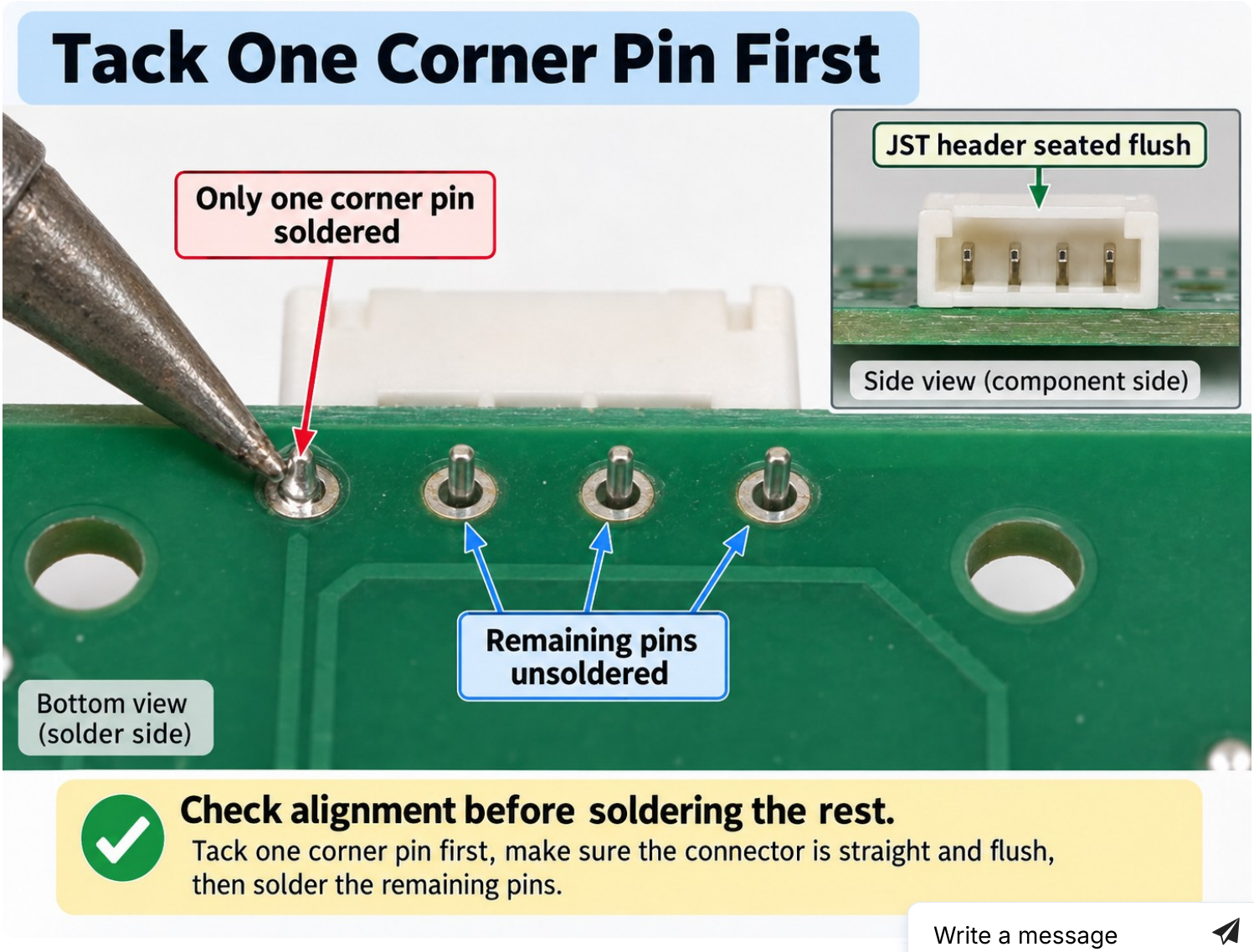
If solder beads up or a large pin does not wet evenly, add flux first. Improve heat transfer with the correct tip before simply increasing iron temperature.

07 Trim, inspect, and clean

Trim excess leads where required. Check for full wetting, adequate
joints, and damage to the plastic housing. Clean residue with IPA w
flux and assembly.

Write a message

For workmanship and acceptance context, IPC publishes J-STD-001 for soldered electrical/electronic assemblies and IPC-A-610 for assembly acceptability criteria.



AI-generated

Write a message

JLCPCB

Designing a Board Around JST Connectors?

Moving from a hand-wired prototype to a proper PCB? Turn the connector footprint into a manufacturable board and review the layout before production.

Claim $11 Coupon

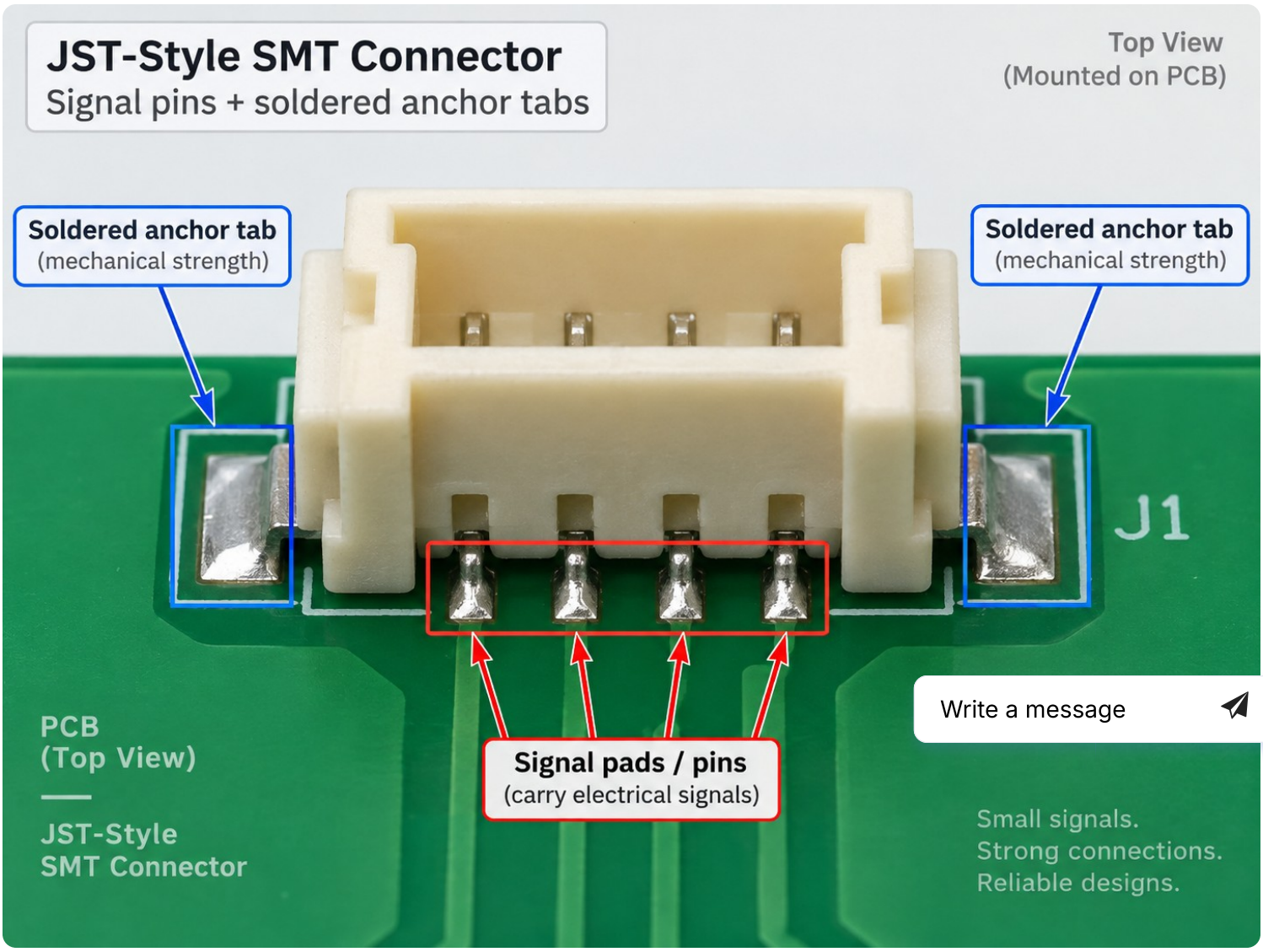
How to Solder an SMT JST Connector

For a surface-mount JST header, start by lightly tinning one pad or one hold-down area. Position the connector with tweezers, reheat that single pad, and let the part settle into the exact footprint location. Once the connector is sitting straight and all pins are properly aligned with the pads, solder the remaining contacts using flux and a fine-tip soldering iron.

Do not treat the signal leads as the only important joints. Many SMT connectors include metal hol down tabs or anchor pads designed to carry insertion force. Solder these completely so the

connector body is mechanically tied to the PCB. Afterward, inspect the fine-pitch pins for bridges, opens, and solder hidden near the body edge.

For repeated small-batch builds, solder paste and reflow are often cleaner and more repeatable than touching every SMT pin by hand. Confirm that the exact JST part number is rated for your reflow process before using that approach.



AI-generated

Should You Solder or Crimp JST Wires?

Use a proper crimp whenever possible. JST wire housings are designed around crimp contacts, and a correct crimp provides electrical contact while preserving conductor flexibility behind the terminal. That flexibility matters because cables move during assembly and use.

Soldering a wire to a contact can be acceptable as a repair or no-tool fallback, but it should not be presented as equivalent to the intended crimp. If solder wicks into the conductor beyond the joint, the wire becomes rigid at the transition point and is more likely to fatigue under repeated bending. Keep any repair joint short, controlled, and mechanically supported.

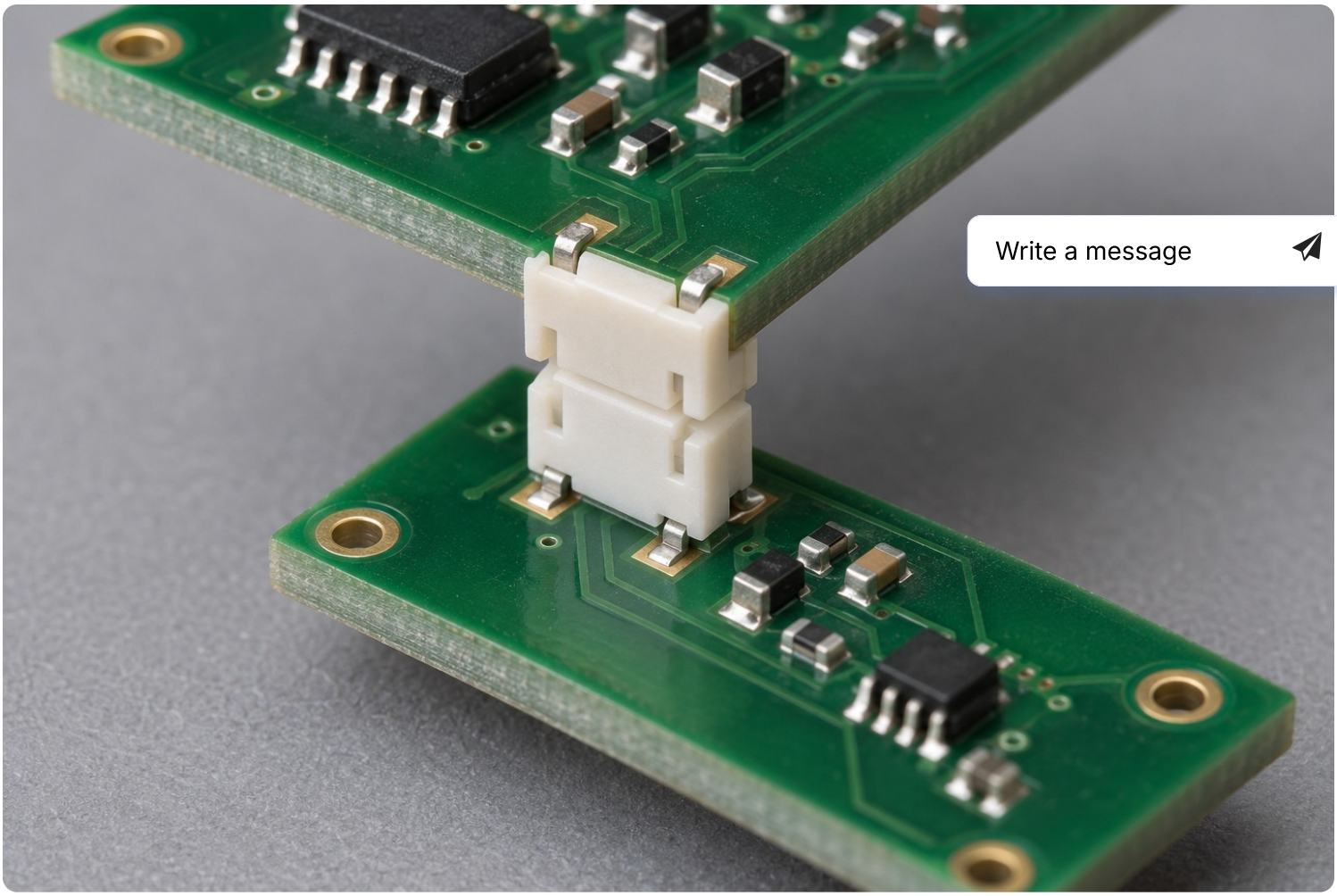
If your real task is attaching wires directly to PCB pads rather than assembling a JST housing, use JLCPCB's separate guide on how to solder wires to a circuit board properly.

How to Solder Board-to-Board JST Connectors

Board to board connector soldering is mainly an alignment problem. The two mating connectors must be coplanar, parallel, and spaced correctly or the boards may not seat even when every joint is electrically sound. Tack one corner first, then verify the connector position before completing the rest.

Test-fit the mating board while only one or two joints are tacked, then remove it before final soldering. Do not solder the pair while mated; heat and fixture stress can distort alignment or damage the connector.

On fine-pitch board-to-board parts, inspect all corners for coplanarity. A lifted corner can leave one row of pins open. Solder any hold-down or shield tabs specified by the connector design; they improve both mechanical strength and, where applicable, shielding continuity.



AI-generated

Can JST Connectors Be Reflow Soldered?

SMT JST Connectors in Reflow

Yes, many SMT JST headers can be reflowed when the exact part is specified for that process. Apply solder paste to the connector pads, place the header with the rest of the SMD parts, and run the board through the approved reflow profile. The benefit is one controlled thermal cycle and repeatable joints across multiple boards.

The connector datasheet defines the allowable soldering and reflow conditions. Housing materials and reflow ratings vary by series and part number, so do not assume every SMT header tolerates the same peak temperature or cycle count.

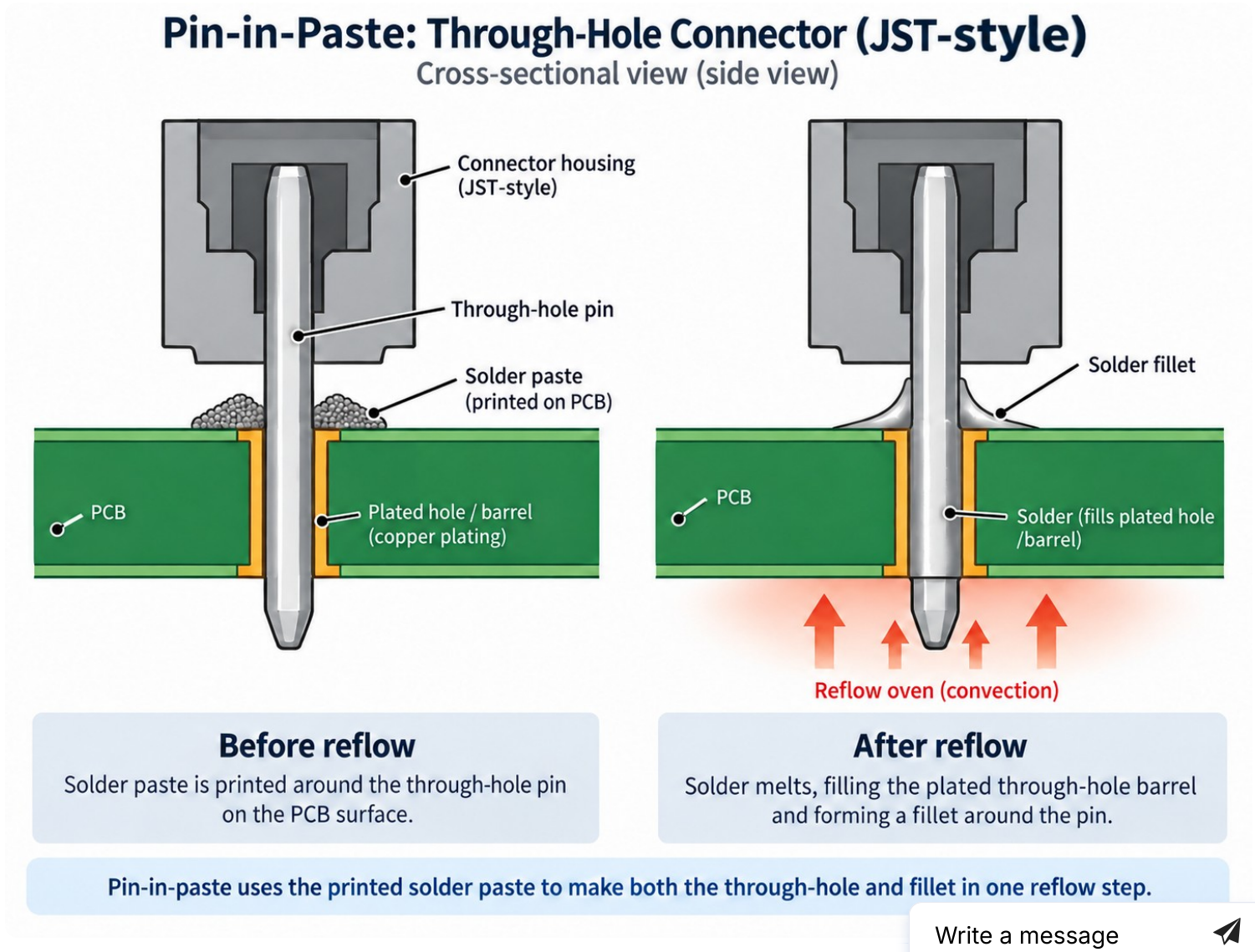
Through-Hole JST Connectors with Pin-in-Paste

Pin-in-paste, also called intrusive reflow, can sometimes solder a through-hole connector during the same reflow cycle as the SMD parts. Solder paste is printed around the through-hole openings, the connector pins are inserted, and the molten paste is drawn into the plated barrels during reflow.

This method is conditional. Suitability depends on the connector design, pin geometry, PCB thickness, hole size, required solder volume, and the manufacturer's reflow specification. Not every through-hole JST connector should be processed this way. Verify the connector and board design before committing to production.

For the detailed process, see JLCPCB's pin-in-paste through-hole reflow with a stencil guide. Paste volume and clean stencil release become especially important when the stencil must feed both ordinary SMD pads and through-hole barrels.

Write a message



AI-generated

Write a message

JLCPCB

Reflowing Connectors? Add a Solder-Paste Stencil

For SMT or pin-in-paste builds, a correctly designed stencil helps control paste volume across signal pads, anchor tabs, and through-hole apertures.

Get an Instant Quote

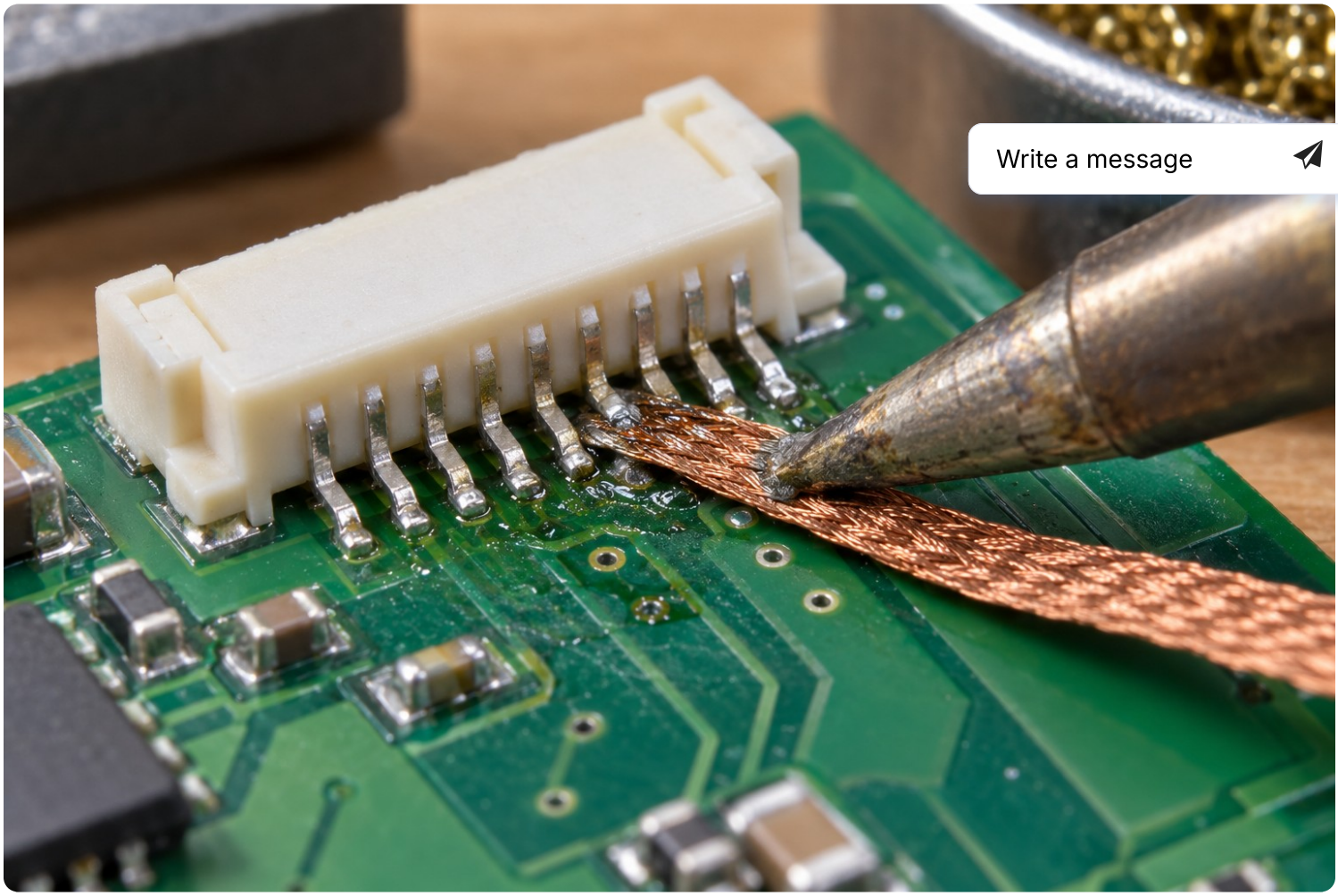
Common JST Connector Soldering Problems and How to Fix Them

- **Problem: Connector sits crooked or lifted off the board.** Fix: Tack one pin first, reheat it, and press the housing flush before soldering the rest. Do not lock a tilted connector in place with multiple completed joints.
- **Problem: Power or ground pins look dull and will not wet.** Fix: Large pins and copper planes p heat away. Use a larger tip, add flux, increase contact time moderately, and preheat the board i

available before raising the set temperature.

- **Problem: Through-hole pins are soldered but the barrels are not filled well.** Fix: Apply enough heat and solder to wet the pin and plated barrel. Inspect both sides where accessible and correct incomplete fill before the connector sees mechanical load.
- **Problem: Solder bridges form between close pins.** Fix: Add flux, then remove excess solder with the iron tip or solder wick. Keep deposits small and inspect under magnification on fine-pitch parts.
- **Problem: A board-to-board pair will not mate.** Fix: Suspect coplanarity or spacing. Reheat a tacked corner, test-fit the mating board, and correct position before finishing all joints.
- **Problem: The connector housing melts or deforms.** Fix: Reduce dwell time and avoid using excessive iron temperature to compensate for poor heat transfer. Use the correct tip, work one pin at a time, and let the housing cool.
- **Problem: The connector pulls off under cable strain.** Fix: Rework the mechanical joints. Solder every hold-down tab, ensure through-hole joints are fully formed, and add strain relief where the product design needs it.

For deeper troubleshooting, use JLCPCB's separate guides on preventing solder bridging and cold solder joints.



AI-generated

FAQ about Soldering JST Connectors

Q: What Soldering Iron Tip Is Best for a JST Connector?

Q: How Hot Should the Iron Be for JST Connector Soldering?

Q: Can I Solder a JST Connector Backward and Fix It Later?

Q: Do JST Connector Anchor Tabs Need Solder?

Q: Can I Use Hot Air to Remove an SMT JST Connector?

Q: Why Does My JST Power Pin Need More Heat Than the Signal Pins?

Q: Should I Hand Solder or Reflow an SMT JST Header?

Write a message

Q: How Can I Make a JST Connector Joint More Mechanically Reliable?

Conclusion: How to Solder a JST Connector Successfully

The reliable workflow is simple: identify the correct JST series and mount type, seat the connector flush, tack one pin, check that it is square, then complete the remaining electrical and mechanical joints. For SMT and board-to-board parts, alignment and coplanarity deserve the same attention as solder quality.

The main lesson is that connector soldering is electrical and mechanical at the same time. Full joints, soldered anchor tabs, proper barrel fill, and strain relief are what keep a connector working after hundreds of insertions and cable movements. For production boards, reflow—and pin-in-paste where the exact connector supports it—can be faster and more repeatable than hand-soldering many pins.

For a new board, verify the exact footprint and datasheet first, check the mechanical fit, then choose hand soldering or reflow based on connector type and build volume. That makes learning how to solder JST connector hardware less error-prone.

Keep Learning

How to Solder a JST Connector: Through-Hole, SMT, and Board-to-Board Guide

Key Takeaways Soldering a connector is electrical and mechanical at once: Seat the header flush and square, tack one corner pin, verify alignment, then solder the remaining pins and every...

How to Tin a Soldering Iron: First Use & Re-Tinning

Key Takeaways Tin from the first heat-up: Bring the iron to roughly 315–370 °C (600–700 °F), wipe the tip on brass wool, and melt fresh flux-cored solder onto every working face until a thin...

DC Jack Soldering: How to Solder a DC Barrel Jack / Power Connector

Key Takeaways Confirm polarity before the iron touches the board: Check the adapter label, device marking, and jack pinout for center-positive versus center-negative wiring, because...

How to Solder an Audio Jack to a PCB

Write a message

Key Takeaways Confirm the exact pinout first: TS, TRS, and TRRS jacks differ in conductor count and PCB terminal numbering, and TRRS wiring follows CTIA or OMTP standards, so always ma...

How to Solder SMD LEDs: A Step-by-Step Guide for Iron, Hot Air, and Reflow

Key Takeaways Confirm polarity and orientation first: Identify the anode and cathode before applying any heat, because a reversed SMD LED will not light and correcting it costs another...

How to Solder a Potentiometer (PCB, Panel & Guitar Pots)

Key Takeaways Identify the wiper terminal: In most potentiometers, the middle pin is the wiper. Always verify with a multimeter before soldering. Pre-tin surfaces: Tin both the terminals and...

© 2026 JLCPCB.COM All Rights Reserved. Privacy Policy Terms & Conditions Cookies Policy

Do Not Sell or Share My Personal Information

Write a message